

HARPREET SINGH

Email: workwithharpreetsingh@gmail.com

LinkedIn: [linkedin.com/in/harpreet-singh-b42942213/](https://www.linkedin.com/in/harpreet-singh-b42942213/)

Research Assistant

M.Sc Computer Science

Mobile : +1 778-536-4298

Location : Lethbridge, AB

PROFESSIONAL SUMMARY

Technology leader with 13+ years of experience spanning electronics design, embedded systems, data engineering, AI/ML, and applied research. Hold a Master's in Computer Science and a Bachelor's equivalent in Electronics & Telecommunication (AMIE/TE, IETE), with a proven track record of delivering end-to-end solutions — from hardware prototyping to secure data pipelines and AI-powered analytics. As a Research Assistant in Neuroscience & AI, I work on decoding brain signals (EEG/fNIRS) for brain-computer interface (BCI) applications using advanced signal processing and machine learning. In cybersecurity research, I design AI-driven methods for anomaly detection, embedded device security, and resilient IoT architectures.

Core Competencies:

- Electronics & Embedded Systems: PCB design, firmware development, drivers, IoT device engineering.
- Data Engineering & Cloud: Data pipelines (SQL, Spark, Databricks, Azure/AWS), schema evolution, large-scale analytics.
- AI & Machine Learning: Edge AI, predictive analytics, biomedical data analysis, TinyML.
- Cybersecurity: Secure boot, OTA updates, cryptographic protocols, AI-enhanced threat detection.
- Research & Innovation: Neuroscience + AI (functional connectivity, BCI), cybersecurity AI applications.
- Leadership: Project lead, multidisciplinary R&D teams, translating research into scalable products.

With a unique blend of hands-on engineering expertise and research-driven innovation, I am well-positioned to drive technology strategy and execution as a CTO or Senior Engineering Leader in IoT, healthcare, AI, and secure systems.

TECHNICAL SKILLS

Embedded Systems & Hardware Design:

- Microcontrollers: ARM Cortex (STM32, LPC), AVR, PIC, 8051, ESP32, Atmega328P.
- Embedded Software Development: Bare Metal, FreeRTOS, Zephyr RTOS, Embedded Linux
- Interface Protocols: UART, SPI, I2C, CAN, RS-485, Modbus, USB, Ethernet, MQTT.
- Circuit Design and PCB Layout: Altium Designer, KiCad, DipTrace, Eagle.
- IoT Technologies: LoRa, Zigbee, Bluetooth (BLE), Wi-Fi, GSM/GPRS, GPS.

Programming & Development:

- Languages: C, Embedded C, Assembly, Python, Bash.
- Development Tools: Keil uVision, MPLAB, STM32CubeIDE, Arduino, CCS C Compiler, Eclipse, Visual Studio Code.
- Debugging & Testing: JTAG, SWD, Logic Analyzers, Oscilloscopes, Wireshark, Zeek, Putty.

Operating System:

- Windows, Linux (Ubuntu, Debian, Kali Linux), MacOS.

Networking & Cybersecurity:

- TCP/IP, UDP, ICMP, MQTT, HTTP, WebSocket, SSL/TLS**
- Secure boot, Cryptography, Secure Firmware Updates, Ethical Hacking.
- Penetration Testing of IoT devices, OSCP (Test Pending).

Libraries and Tools:

- Numpy, Scikit-learn, TensorFlow, Pandas, Matplotlib, PyTorch, OpenCV, Django.

Development Tools:

- Git, Anaconda, Pycharm, Putty, WinSCP, VMware, Weka, Zeek, Wireshark, Blender.

WORK EXPERIENCE

Core-Hub Neuroengineering Solutions at University Of Lethbridge (Research Assistant)

April 2025- Present

- Contributes to ongoing innovation in applied neuroengineering and embedded systems. My work spans multiple domains, including:
- Development of brain-computer interface technologies and wireless implantable systems.
- Signal processing and analysis of neurophysiological data such as EEG and fNIRS.
- Hardware and software development for embedded RFID-based monitoring solutions in agricultural and health-tech environments.
- Collaborative research with Brain Care Technology (BCT) and HerdLinx RFID Solutions as part of Core-Hub's mandate to translate neuroscience discoveries into real-world applications.
- Project: Brain Computer Interface: 8-Channel EEG, 8-Channel fNIRS, 10-Source Led IR emitter
- Making the complete product involving designing of Embedded device, Embedded Software, Firmware and setting CI/CD pipeline. Aftware the data acquisition, data analysis and visualization using python based tool like matplotlib, seaborn etc

Independent Embedded Systems Developer (Self-Employed)

June 2024 - Mar 2025

- **Project: PyBoard.**
 - **Microcontroller:** STM32F405RGT6(Main controller).
 - **Role in the Project:** This project involved designing and developing a custom Pyboard, a microcontroller-based development board that runs MicroPython, enabling Python scripting for embedded applications. The board is built around an STM32 microcontroller, featuring various communication interfaces, GPIOs, and power management circuits for efficient operation.
 - **Description:** The project involved designing a multi-layer PCB, flashing MicroPython firmware, and integrating support for sensors, actuators, and communication modules. I optimized power management and debugged hardware/software interactions, making the board suitable for applications such as IoT prototyping, embedded system control, automation, robotics, and cybersecurity monitoring. This project demonstrates my expertise in PCB design, firmware development, and MicroPython integration

University of Regina (Research Assistant)

Jan 2021- Apr 2024

Research: A Robust Intrusion Detection System Utilizing Uncertain Reasoning Techniques in Artificial Intelligence.

- Developed an advanced Intrusion Detection System (IDS) leveraging probabilistic reasoning techniques to detect and mitigate DoS/DDoS attacks.
- Implemented AI models using Bayesian Networks and Markov Networks, achieving high detection accuracy with minimal false positives.
- Analyzed network traffic data using Python, TensorFlow, and PyTorch for IDS training and validation.
- Employed tools like Zeek, Wireshark, and Scikit-learn for data collection, preprocessing, and feature extraction, enhancing system performance.
- Implemented feature engineering and model optimization using Grid Search and K-Fold Cross Validation.
- Validated system effectiveness through real-world testing, strengthening network security strategies.
- Presented findings at academic conferences and collaborated with faculty advisors for further research refinement.

Research: An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection

- Published research in the 21st International Conference on Security and Cryptography (SECRYPT 2024), ISBN 978-989-758-709-2, ISSN 2184-7711, pages 771-776.
- Proposed an IDS model leveraging Bayesian Networks and Markov Networks to improve DoS/DDoS detection and mitigation.
- Conducted experiments using Python, TensorFlow, PyTorch, Zeek, and Wireshark, validating the system's efficiency.
- Engaged with global cybersecurity professionals through conference presentations and discussions.

Project: A Bayesian Network Model for Diagnosis of Liver Disorders

- Implemented probabilistic graphical models for medical diagnosis using Bayesian Networks.
- Developed a Python-based diagnostic tool leveraging pgmpy, networkx, Pandas, and Scikit-learn.
- Designed an ML-powered medical inference system for clinical decision-making, achieving 85% accuracy.

Project: Punjabi BERT- Natural Language Processing Model

- Built a deep learning-based NLP model using Transformer architecture, improving classification accuracy for over 100K+ news articles by 92%.
- Preprocessed text data using tokenization, stop word removal, and lemmatization with NLTK and Keras.
- Conducted transfer learning and hyper-parameter tuning, optimizing Transformer performance with K-Fold Cross-Validation.
- Analyzed large Punjabi text corpora using Python, TensorFlow, PyTorch, and OCR techniques for effective text processing.

Planetcast Media Services Ltd (Senior Software Engineer) *

Dec 2018 - Nov 2020

- Led a 15-member team in designing and developing embedded devices and system software for real-time data acquisition and processing.
- **Project:** Network based 10 channel Video Switcher.
 - **Microcontroller:** ATMega328P.
 - **Role in the Project:** Responsible for creating the complete schematic diagram, designing the PCB using Diptrace, and selecting new components for the project. ■ Collaborated with the mechanical department to discuss the mechanical structure for both the PCB and the system-level cabinet. ■ Developed the complete firmware in C language.
 - **Description:** This project is used for switching the channel from main server to backup server and vice-versa, via command sent on network. It can also be controlled manually by a Desk control unit.
- **Project:** Ethernet based Temperature Acquisition System.
 - **Microcontroller:** ATMega328P and Atmega2560A.
 - **Role in the Project:** I was responsible for designing the complete schematic diagram, developing the PCB layout in Diptrace, and selecting the new components to be used. I also collaborated with the mechanical department to discuss the mechanical structure of the PCB and system-level cabinet. Additionally, I developed the full firmware for the HTTP server using LWIP (Light Weight Internet Protocol) in C language and created multiple web pages as a front-end application to control hardware and monitor sensors.
 - **Description:** This project is used in Various Industrial application where there is a need to check the temperature of the particular area where human beings are restricted to go and it also provide Front end application for the users.

- **Project:** Ethernet Switcher.
 - **Microcontroller:** ATmega328P.
 - **Role in the Project:** I created the complete schematic diagram, designed the PCB using Diptrace, and selected new components for the project. I coordinated with the mechanical department to discuss the mechanical structure for both the PCB and the system-level cabinet. Additionally, I developed the entire firmware in C language.
 - **Description:** This project is used to Switch between two Ethernet input (channel A and channel B) coming from two different servers (Main and backup). Selected channel can output its data to four separate channels. This type of Device is used in media industry where we have Main Server and backup server. If there is a problem in Main server then this unit will automatically switch to backup server and connects the Video and audio streaming channel to the 4 output channels and generate alarms accordingly if any server gets down.

Spark Eighteen Pvt. Ltd. (Embedded Software Engineer) *

March 2018 - Dec 2018

- **Project:** Ethernet controlled LIFI Network .
 - **Microcontroller:** STM32F429ZIT6(Main controller).
 - **Role in the Project:** I created the complete schematic diagram and assisted the PCB designer in selecting appropriate footprints for new components. I also designed a 15W Switch Mode Power Supply (SMPS) to power various peripherals on the PCB, with an input voltage range of 20Vdc to 80Vdc and output voltages of 3.3V and 5V (isolated), achieving an efficiency of 79%. Additionally, I developed the full firmware for the HTTP server using LWIP (Lightweight Internet Protocol) and created multiple web pages for the front-end interface to control the hardware.
 - **Description:** This project is used to provide Front end application for the user who have LIFI internet systems in their work environment to control the intensity of light and to Block/Unblock the data service from that particular light source.

Exicom Tele-systems Ltd. (R&D Engineer) *

Feb 2016 - Feb 2018

- **Project:** Power Plant Unit.
 - **Microcontroller:** STM32F427ZIT6(Main controller) and STM32F030CT6(Analog controller).
 - **Role in the Project:** I was responsible for designing the complete schematic diagram and assisting the PCB designer in selecting suitable footprints for new components. Additionally, I designed a 15W Switch Mode Power Supply (SMPS) to power various peripherals on the PCB, with an input voltage range of 32Vdc to 72Vdc and output voltages of 3.3V (main control voltage), 3.3V (analog), isolated 5V, and unregulated 12V, achieving an efficiency of 81%. I also collaborated with the mechanical department to discuss the mechanical structure of the PCB and the system-level cabinet. Furthermore, I facilitated the board bring-up by developing simple test software for different sections of the PCB, including I2C protocol-based EEPROM and SPI protocol-based Flash.
 - **Description:** This project is used to provide continues power supply (Telecom standard 48V dc) to the Reliance Jio WIFI Hotspot site.
- **Project:** DCIF Card (DC Interface Card) with 6-Voltage and 7-Current channels.
 - **Microcontroller:** STM32F373VCT6.
 - **Role in the Project:** I was responsible for designing the complete schematic diagram and assisting the PCB designer in selecting appropriate footprints for new components. Additionally, I designed a 15W Switch Mode Power Supply (SMPS) to power various peripherals on the PCB, with an input voltage range of 22Vdc to 72Vdc and output voltages of +15V (main control voltage), 3.3V, isolated 5V, and -15V, achieving an efficiency of 78%. Furthermore, I contributed to the board bring-up process by developing firmware for testing purposes.
 - **Description:** This type of card is used to sense and measure DC main Supply voltage (48V telecom standard voltage), Battery Voltage, Battery current, load current, Temperature and send data to Main controller through CAN communication.
- **Project:** DCIF Card (DC Interface Card) with 4-Voltage and 2-Current channels.
 - **Microcontroller:** STM32F373RCT6.
 - **Role in the Project:** I was responsible for creating the complete schematic diagram and assisting the PCB designer in selecting suitable footprints for new components. Additionally, I designed a 20W Switch Mode Power Supply (SMPS) to power various peripherals on the PCB, with an input voltage range of 22Vdc to 72Vdc and output voltages of +15V (main control voltage), 3.3V, isolated 5V, and -15V, achieving an efficiency of 78%. Furthermore, I contributed to the board bring-up process by developing firmware for testing purposes.
 - **Description:** This type of card is used to sense and measure DC main Supply voltage (48V telecom standard voltage), Battery Voltage, Battery current, load current, Temperature and send data to Main controller through CAN communication.
- **Project Name:** Fly-back Switched Mode Power Supply (SMPS).
 - **Switching IC:** LM5020.
 - **Role in the Project:** • I was responsible for creating the complete schematic diagram and assisting the PCB designer in developing the PCB layout. Additionally, I designed the Flyback transformer and provided guidance to the technician during the winding process.
 - **Description:** This type of power Supplies can be used to make different level of Desired DC to DC voltages.
- **Project Name:** Automatic Test equipment (ATE) to Test NMS (Network monitoring system) card.
 - **Microprocessor Used:** General purpose Computer.
 - **Programming Language Used:** Python, Interpreter: Python 2.7
 - **Role in the Project:** To make complete software application with Graphical User Interface (GUI) to test NMS card.
 - **Description:** This type of Software application is used to test GSM hardware module by providing the Valid set of AT commands.

EDUCATION

University of Regina (Graduated in 2024)

MS in Computer Science (Thesis)

Specialization: Data Engineering, Data Analysis, Artificial Intelligence, Research

Classes focused on Artificial Intelligence, Data Science, Cybersecurity, Cryptography, Advanced Data Structure Designing, Operating System.

Institute Of Electronics and Telecommunication Engineer (Graduated in 2012)

Bachelors in Electronics & Telecommunication Engineering (AMIETE, IETE)

Specialization: Full Stack Embedded system development, Real-Time Operating System.

Classes focused on Embedded system designing, Real-Time Operating System, Microcontroller and Microprocessors, Signals and Systems, Control system engineering, Electromagnetic Radiation, C and Data Structures

EXTRA CURRICULARS

- Worked with director of JABC to gather more volunteers for the providing mentorship to the young leaders in the school.
- Serve the Sikh Community by doing chores at different Sikh community temples.
- Provided coaching and mentorship sessions for young students joined university regarding mental and physical well-being on top of studies.